

AERIS

Production Run, Post-Run QC & Debug Reference

1 — Massive Unattended Production Run

A full 100 k-geometry aero dataset generated overnight with LHS sampling. Plots are disabled for speed. AeroSandbox is kept on so the AVL solver can reconstruct geometry from disk. The **--qc-preset production** flag bundles geometry + aero QC with the *strict* profile and *fail-on-error* behaviour.

```
aeris dataset aero-generate \  
-c configs/geometry/baseline_bwb_25.yaml \  
--n 100000 \  
--sampler lhs_v1 \  
--sampler-seed 123 \  
--name bwb_ai_prod_v1 \  
--no-save-plot \  
--build-aerosandbox \  
--alpha-values 0,2,4 \  
--velocity-values 28 \  
--altitude-values 1500 \  
--control-input-values -5,0,5 \  
--avl-command avl \  
--keep-geometry-dataset \  
--qc-preset production
```

Flag breakdown

Flag	Value	Purpose
--n 100000	100 000	Number of LHS geometry samples to generate and evaluate
--sampler lhs_v1	lhs_v1	Latin Hypercube Sampling — maximally space-filling
--sampler-seed 123	123	Reproducible sampling across runs
--name bwb_ai_prod_v1	string	Output dataset folder under data/datasets/
--no-save-plot	flag	Skip per-geometry planform PNG — critical for speed at scale
--build-aerosandbox	flag	Build AeroSandbox airplane object (needed by AVL solver)
--alpha-values 0,2,4	deg	Angle-of-attack sweep: 3 points × 100 k geoms = 300 k cases
--velocity-values 28	m/s	Single cruise velocity
--altitude-values 1500	m	Single cruise altitude
--control-input-values -5,0,5	deg	Elevon deflection sweep — 3 values, 0 is the neutral ref
--avl-command avl	avl	AVL executable name; must be on PATH
--keep-geometry-dataset	flag	Preserve intermediate geometry dataset for reproducibility
--qc-preset production	preset	Enables strict geometry + aero QC with fail-on-error (to be impl.)

■ `--qc-preset production` is a proposed flag. In the current codebase it maps to: `--run-geometry-qc --geometry-qc-profile strict --fail-on-geometry-qc-error --run-aero-qc --aero-qc-profile strict --fail-on-aero-qc-error`. Add the preset resolver to `commands/dataset.py` before using it.

■ Total sweep size: $100\,000 \text{ geoms} \times 3 \text{ alpha} \times 1 \text{ velocity} \times 1 \text{ altitude} \times 3 \text{ control} = 900\,000 \text{ AVL solver calls}$. Estimated wall time at $\sim 0.5 \text{ s/call}$ with AVL: $\sim 125 \text{ hrs}$ single-threaded. Use a process pool or cluster for production.

2 — Post-Run Curation / Acceptance

Run strict QC post-hoc on the completed datasets. The geometry dataset lives under `bwb_ai_prod_v1__geometry` (intermediate), the final aero dataset under `bwb_ai_prod_v1`. Either can be inspected independently.

Geometry dataset QC

```
aeris dataset qc \  
--dataset data/datasets/bwb_ai_prod_v1__geometry \  
--profile strict
```

Aero dataset QC

```
aeris dataset aero-qc \  
--dataset data/datasets/bwb_ai_prod_v1 \  
--profile strict
```

What strict profile checks

Flag	Value	Purpose
geometry (basic)	always	manifest consistency, required files, duplicate IDs, NaN values, basic ranges
geometry (strict)	+above	same as basic (strict == basic in current profiles.py)
aero (basic)	always	manifest consistency, finite CL/CD/Cm, grid completeness, zero-ctrl ref, control diagnostics, control effectiveness, basic ranges
aero (strict)	+above	CL-alpha monotonicity, Cm-control trend, CD outlier scan

■ Both commands exit nonzero if QC fails when called standalone — the exit code propagates correctly to shell scripts and CI runners. QC reports are written to `/qc/geometry_qc_report.json` and `aero_qc_report.json`.

3 — Debug Run

A minimal smoke-test run to validate the pipeline end-to-end before committing to a production job. Use `--qc-preset debug` (proposed) for lightweight, non-failing QC that still surfaces warnings.

```
aeris dataset aero-generate \
-c configs/geometry/baseline_bwb_25.yaml \
--n 5 \
--sampler lhs_v1 \
--sampler-seed 42 \
--name bwb_debug_run \
--save-plot \
--build-aerosandbox \
--alpha-values 0,2 \
--velocity-values 28 \
--altitude-values 1500 \
--control-input-values -5,0,5 \
--avl-command avl \
--qc-preset debug
```

Debug vs production preset comparison

Flag	Value	Purpose
--qc-preset debug	proposed	basic QC, no fail-on-error, warnings printed to stdout
--qc-preset production	proposed	strict QC, fail-on-error on both geometry and aero
(no preset)	default	no QC — fastest, use only if QC runs separately post-hoc
--run-qc --qc-profile basic	explicit	equivalent to debug preset for geometry-only dataset

■ For the debug run use `configs/smoke/dev.yaml` or `baseline_bwb_25.yaml` (the `_25` variant uses `n_spline=10/5` — much faster than the full `150/50` variant). Swap to `configs/geometry/wing_bwb.yaml` only when the pipeline is validated.

4 — Implementing --qc-preset (Required)

The `--qc-preset` flag does not exist yet. Add a simple resolver in `src/aeris/commands/dataset.py` before the `run_aero_dataset_generation()` call:

```
# In dataset_aero_generate() before the run call:
QC_PRESETS = {
    "production": dict(
        run_geometry_qc=True, geometry_qc_profile="strict",
        fail_on_geometry_qc_error=True,
        run_aero_qc=True, aero_qc_profile="strict",
        fail_on_aero_qc_error=True,
    ),
    "debug": dict(
        run_geometry_qc=True, geometry_qc_profile="basic",
        fail_on_geometry_qc_error=False,
        run_aero_qc=True, aero_qc_profile="basic",
        fail_on_aero_qc_error=False,
    ),
}

if qc_preset:
    preset_kwargs = QC_PRESETS.get(qc_preset, {})
    run_geometry_qc = preset_kwargs.get("run_geometry_qc", run_geometry_qc)
    geometry_qc_profile = preset_kwargs.get("geometry_qc_profile", geometry_qc_profile)
    fail_on_geometry_qc_error = preset_kwargs.get("fail_on_geometry_qc_error", ...)
    run_aero_qc = preset_kwargs.get("run_aero_qc", run_aero_qc)
    aero_qc_profile = preset_kwargs.get("aero_qc_profile", aero_qc_profile)
    fail_on_aero_qc_error = preset_kwargs.get("fail_on_aero_qc_error", ...)
```

